B E N C H M A R K R E P O R T

Same Answer, Different Journeys

When 1 Token and 3,000 Tokens Both Get It Right

Behavioral Analysis of 22 LLMs on a Wastewater Sample Preservation Question

22	95.5%	3,000×	5th
LLMs Tested	Reported Pass Rate	Token Spread	R1 Parsing Issue

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1. Executive Summary

I posed a laboratory sample preservation question to 22 LLMs: an illegal discharge has entered a wastewater treatment plant in Adana during a July heatwave. The lab technician is out of town until Monday. Field personnel need instructions on how to store the sample. Five options range from negligent (leave at room temperature) to correct (+4°C, closed cap, dark conditions), with a designed trap (acid preservation) targeting models that apply specific-parameter logic to an unknown sample.

The reported pass rate is 95.5% (21/22). The adjusted rate, accounting for DeepSeek R1's parsing issue, is 100% (22/22). This is the seventh benchmark in the WWTP series, and it reveals that when the question is straightforward, the analytical value shifts entirely to behavioral dimensions: how models explain, how much they write, and how they handle a trap that none of them fell for.

F1 — One Token, One Correct Answer: Gemini 2.5 Flash responded with a single character: "D". No explanation, no reasoning, no context. One token. Correct. This is the most minimal response in the entire WWTP benchmark series.

F2 — 3,000× Token Spread for the Same Answer: Qwen3 Next 80B Thinking used 3,009 tokens to arrive at the same correct answer. The spread between the shortest and longest correct response is approximately 3,000× — the widest token efficiency gap observed in the series.

F3 — R1's Fifth Parsing Failure: DeepSeek R1 answered D (correct) inside a <think> block. The parser stripped non-letter characters and read "T" (from "think") as the answer. This is R1's 5th parsing issue in 7 benchmarks. The pattern has now occurred in 71% of all benchmarks.

F4 — The Acid Trap That Caught No One: Option C (add sulfuric acid) was designed as the primary trap — acid preservation is correct for specific parameters but wrong for an unknown sample. 19 of 22 models explicitly discussed this option in their reasoning and rejected it. The trap was recognized and avoided by nearly every model.

F5 — Gemma 1B's Recovery: In the previous benchmark (Confined Space), Gemma 1B gave the most dangerous answer in the series — recommending unprotected entry into a confined space. Here, with 95 tokens, it answered D correctly. The smallest model in the set appears unpredictable rather than consistently wrong.

2. The Question

This benchmark tests basic laboratory knowledge: how to store a wastewater sample when analysis must be delayed. The scenario is grounded in a real operational situation at Ceyhan WWTP in Adana, Turkey — a hot climate where improper storage can degrade a sample within hours.

The question is designed with one clear correct answer (D: +4°C, closed cap, dark conditions) and one deliberate trap (C: acid preservation). Acid addition is a legitimate preservation technique for specific parameters like metals analysis, but for an unknown illegal discharge where comprehensive testing is needed, it would interfere with multiple analytical methods. The remaining options (A, B, E) represent varying degrees of negligence.

With 5 options, the random correct answer probability is 20% — lower than the previous four-option benchmarks (25%) but higher than the Confined Space benchmark's 6.25% (16 options).

Table 1: The Five Options — From Negligent to Correct

Option	Level	Description
A	Negligent	Leave at room temperature. In Adana July (35°C+), rapid bacterial growth and chemical degradation.
B	Harmful	Open cap = contamination, evaporation, volatile compound loss. Accelerates degradation.
C	TRAP	Acid preservation — correct for specific parameters (metals) but destroys unknown sample integrity for general analysis.
D	CORRECT	+4°C refrigeration, closed cap, dark conditions. APHA standard for general preservation. Maintains sample integrity 48–72 hours.
E	Poor	+25°C with light = warm temperature accelerates decay, light triggers photochemical reactions.

Table 1: Option C is the designed trap. Acid preservation is a real technique but inappropriate for unknown samples requiring comprehensive analysis.

3. Results

Figure 1: Answer Distribution

D (Correct)	21 models (incl. R1 parsed wrong)
T (R1 parsing)	1 model (DeepSeek R1 — actual answer D)

Figure 1: Near-unanimous agreement. Every model selected D as the correct answer. R1's parsing issue is the only deviation on record.

Table 2: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Family
Claude Haiku 4.5	D	PASS	295	4.4s	Claude
Claude Opus 4.1	D	PASS	232	10.5s	Claude
Claude Opus 4.5	D	PASS	334	9.7s	Claude
Claude Sonnet 4	D	PASS	217	6.2s	Claude
Claude Sonnet 4.5	D	PASS	355	10.5s	Claude
Gemini 2.0 Flash	D	PASS	338	3.8s	Gemini
Gemini 2.0 Flash Lite	D	PASS	343	4.4s	Gemini

Gemini 2.5 Flash	D	**PASS**	1	8.0s	Gemini
Gemini 2.5 Pro	D	**PASS**	368	17.0s	Gemini
Gemini 3 Flash Preview	D	**PASS**	256	7.0s	Gemini
Gemini 3 Pro Preview	D	**PASS**	297	17.1s	Gemini
Gemma 3 27B	D	**PASS**	~380	8.9s	Gemma
Gemma 3 12B	D	**PASS**	~320	12.5s	Gemma
Gemma 3 4B	D	**PASS**	~300	7.5s	Gemma
Gemma 3 1B	D	**PASS**	95	2.9s	Gemma
DeepSeek R1 (CoT)	D†	**FAIL**	820	6.2s	DeepSeek
DeepSeek V3.1	D	**PASS**	226	2.5s	DeepSeek
DeepSeek V3.2	D	**PASS**	220	11.8s	DeepSeek
Qwen3 235B A22B Instruct	D	**PASS**	232	3.0s	Qwen
Qwen3 Coder 480B	D	**PASS**	208	2.7s	Qwen
Qwen3 Next 80B Instruct	D	**PASS**	237	2.2s	Qwen
Qwen3 Next 80B Thinking	D	**PASS**	3,009	16.1s	Qwen

Table 2: †R1 answered D inside a <think> block. Parser read "T" (from "think") as first character. This is R1's 5th parsing issue in 7 benchmarks.

A note on the single failure: the Kaggle Benchmarks platform evaluates answers by extracting the first alphabetic character from each model's raw response. This is a strict, uniform rule applied across all models. When R1 prepends a <think> block, the first alphabetic character becomes "T" rather than the intended answer. The reported pass rate (95.5%) reflects this rule; the adjusted rate (100%) reflects the actual answer content.

4. LLM Behavior Analysis

4.1 Token Efficiency — 3,000× Spread for the Same Correct Answer

When every model gets the answer right, the most revealing dimension becomes how they get there. On this benchmark, the token spread between the shortest and longest correct responses is approximately 3,000× — the widest gap observed in the WWTP series.

Gemini 2.5 Flash responded with a single token: "D". No explanation, no justification, no discussion of alternatives. One character, correct, done. At the other extreme, Qwen3 Next 80B Thinking used 3,009 tokens for the same letter — including deliberation, elimination of alternatives, and detailed justification.

Between these extremes, a clear efficiency gradient emerges. The compact responders (Gemma 1B at 95 tokens, Qwen Coder at 208, Claude Sonnet 4 at 217) provide concise but adequate reasoning. The mid-range responders (Claude Opus 4.5 at 334, Gemini 2.0 Flash at 338) offer more detailed analysis. The verbose responders (R1 at 820, Qwen Thinking at 3,009) invest substantially more resources for the same outcome.

Figure 2: Token Distribution Across 22 Models

Token Range	Distribution	Count
1 token	Gemini 2.5 Flash	1
1–100	Gemma 1B	1
101–250	Sonnet 4, V3.2, V3.1, Opus 4.1, Qwen 235B, Next 80B Inst	6
251–400	Majority cluster: Claude, Gemini, Gemma, Qwen models	11
401–1,000	DeepSeek R1 (820 tokens incl. think block)	2
1,000+	Qwen Thinking (3,009)	1

Figure 2: Most models cluster in the 251–400 token range. The extremes — 1 token and 3,009 tokens — are both correct, suggesting that token count on this question reflects communication style rather than reasoning necessity.

This pattern is consistent with the Material Selection benchmark (#1), which also had a high pass rate and showed significant token spread. When the cognitive challenge is low, behavioral differences in verbosity, explanation style, and resource allocation become the primary signal.

A latency observation adds nuance: Gemini 2.5 Flash produced its single token in 8.0 seconds, while Qwen3 Next 80B Instruct generated 237 tokens in 2.2 seconds. The fastest response was not the shortest, and the shortest was not the fastest. This suggests that latency on this platform reflects API overhead, model loading, and infrastructure factors in addition to generation time — token count alone is not a reliable proxy for response speed.

4.2 Reasoning Depth — Four Dimensions of a Correct Answer

A correct answer to this question involves understanding four key principles: temperature control (slowing bacterial activity), cap closure (preventing contamination and volatile loss), dark conditions (inhibiting photochemical reactions), and the acid trap rejection (acid preservation is inappropriate for unknown samples). Models varied significantly in how many of these dimensions they addressed.

Table 3: Reasoning Dimension Coverage

Model (selected)	Temp.	Volatile	Dark	Acid Trap	Score
Claude Opus 4.5	✓	✓	✓	✓	**4/4**
Claude Sonnet 4.5	✓	✓	✓	✓	**4/4**
DeepSeek V3.1	✓	✓	✓	✓	**4/4**
Gemini 2.5 Pro	✓	✓	✓	✓	**4/4**
Qwen3 235B	✓	✓	✓	✓	**4/4**
Qwen3 Next 80B Inst	✓	✓	✓	✓	**4/4**
Gemma 3 27B	✓	✓	✓	✓	**4/4**

Claude Opus 4.1	✓	✓	✓	✗	**3/4**
Claude Sonnet 4	✓	✓	✓	✗	**3/4**
Gemini 2.0 Flash	✓	✗	✓	✓	**3/4**
Gemma 3 12B	✓	✗	✓	✓	**3/4**
Gemma 3 4B	✓	✗	✗	✗	**2/4**
Gemma 3 1B	✓	✗	✗	✗	**2/4**
Gemini 2.5 Flash	✗	✗	✗	✗	**0/4**

Table 3: Selected models showing the reasoning depth gradient. All answered correctly, but their explanations ranged from zero dimensions (Gemini 2.5 Flash) to all four. Larger models tend to provide more dimensions, though exceptions exist.

The reasoning depth gradient suggests that even when accuracy is uniform, the quality of justification varies substantially. A model that answers D and explains all four preservation principles provides more value in a real advisory context than one that simply outputs the letter. However, on a benchmark, both receive the same score.

4.3 R1's Parsing Pattern — Fifth Occurrence in Seven Benchmarks

DeepSeek R1's response begins with <think> and contains a 2,856-character deliberation before outputting "D" as its answer. The parser's regex strips non-letter characters from the start, encounters the word "think" from the <think> tag, and reads "T" as the first letter. The result: a correct answer scored as a failure.

This is the fifth time R1 has encountered a parsing issue in seven benchmarks. The pattern is now present in 71% of all WWTP benchmarks:

Table 4: R1 Parsing History Across the WWTP Series

Benchmark	Raw Char	Answer	Correct?	Issue
#1 Material Selection	<	B	Yes (PASS)	<think> block
#2 Pump RCA	<	A	Yes (FAIL*)	<think> block → parsed wrong
#3 Dewatering RCA	<	A	No (FAIL)	<think> block + wrong answer
#4 Biogas	—	—	—	(data pending)
#5 Walkway Material	*	B	No (FAIL)	**B** markdown
#6 Confined Space	*	O	Yes (FAIL*)	**O** markdown → parsed wrong
#7 Sample Preservation	<	D	Yes (FAIL*)	<think> block → T parsed

Table 4: R1 has encountered parsing issues in 5 of 7 benchmarks. In 3 cases, the correct answer was lost to formatting. The <think> block pattern from early benchmarks has reappeared after two benchmarks of bold markdown.

An additional observation: this benchmark marks a return to the <think> block pattern after R1 had shifted to bold markdown (**X**) in benchmarks 5 and 6. This suggests R1's formatting behavior may alternate between these two modes rather than following a linear evolution.

Two other models also used bold markdown on this benchmark: Claude Haiku 4.5 (**D**) and Gemma 3 27B (**D**). In both cases, the parser correctly extracted D. The parsing fragility appears specific to R1's <think> block, where the tag content itself introduces a spurious letter.

4.4 The Acid Trap — Designed, Discussed, and Dismissed

Option C (add sulfuric acid to the sample) was designed as the primary trap for this benchmark. Acid preservation is a legitimate laboratory technique — APHA Standard Methods prescribes it for specific parameters such as metals analysis. The trap tests whether models can distinguish between parameter-specific preservation and general preservation for an unknown sample.

The result: no model fell for it. Of the 22 models, 19 explicitly discussed option C in their reasoning and rejected it. The most common rejection rationale was that acid addition would interfere with other analytical methods, making comprehensive analysis impossible. Three models (Gemini 2.5 Flash, Gemma 4B, Gemma 1B) did not discuss C at all — but also did not select it.

This is the first benchmark in the WWTP series where the designed trap caught zero models. In previous benchmarks, the trap was effective: the Aerobic Trap in Biogas caught 85.7% of models, the Corrosion Trap in Walkway caught 59%, and the "Almost Safe" trap in Confined Space caught 18%. The acid preservation question appears to be below the difficulty threshold of the current model set.

Table 5: Updated Trap Taxonomy — Seven Benchmarks

Benchmark	Trap Name	The Shortcut	Pass %	What Was Missed
#1 Material Selection	—	—	100%	—
#2 Pump RCA	Electrical Trap	High current = winding damage	77%	Mechanical overload
#3 Dewatering RCA	Local Trap	Vibration = bearings	68%	Upstream grit system
#4 Biogas	Aerobic Trap	SOB = aerobic → aeration	14%	Digester sulfur cycling
#5 Walkway	Corrosion Trap	H_2S = best corrosion resist.	36%	Brittle failure at height
#6 Confined Space	"Almost Safe"	Strong protocol = sufficient	68%	Engineer's presence
#7 Sample Preservation	Acid Trap	Acid preserv. = always correct	95.5%	Unknown sample → general preserv.

Table 5: Seven benchmarks, seven different cognitive challenges. The acid trap is the first in the series to catch zero models, suggesting this type of laboratory knowledge is well-represented in current LLM training data.

4.5 Cross-Benchmark Context — A Second "Easy" Question

This benchmark's adjusted 100% pass rate places it alongside Material Selection (#1) as the second question where every model answered correctly. The parallel is instructive: both are relatively straightforward domain knowledge questions, and in both cases the analytical value lies not in accuracy but in behavioral variation.

Material Selection (#1) revealed an 8× token spread across models answering the same correct question. Sample Preservation (#7) expands this to approximately 3,000× — driven primarily by Gemini 2.5 Flash's 1-token response, which had no equivalent in the earlier benchmark.

The difficulty gradient across seven benchmarks now spans from 14% (Biogas) to 100% adjusted (Material Selection and Sample Preservation). This range confirms that the series is probing genuinely different levels of domain challenge, and that LLM behavior varies more between questions than between models on easy questions.

5. Cross-Benchmark Comparison — Seven Questions Deep

Figure 3: Pass Rate Across 7 WWTP Benchmarks

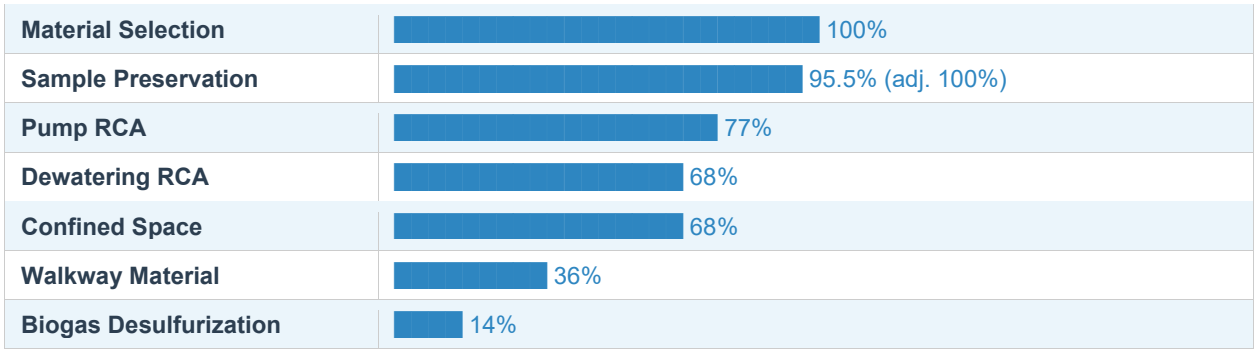

Figure 3: Seven benchmarks spanning 14% to 100%. The Sample Preservation benchmark joins Material Selection at the top of the difficulty gradient, confirming that basic laboratory knowledge is well within current LLM capabilities.

The series now covers seven distinct cognitive dimensions: basic knowledge (Material Selection), symptom elimination (Pump RCA), upstream reasoning (Dewatering), trade-off weighing (Walkway), domain depth (Biogas), safety procedure completeness (Confined Space), and laboratory protocol knowledge (Sample Preservation). No single capability predicts performance across all dimensions.

The cumulative funnel remains at zero. After seven benchmarks, no model has passed every question in the series. The funnel progression: 22 → 17 → 15 → 2 → 0 → 0 → 0. The two easy benchmarks (#1 and #7) did not reopen the funnel — the damage was done by benchmarks 4 and 5, where pass rates of 14% and 36% eliminated all remaining perfect records.

6. Conclusions

This benchmark adds a seventh dimension to the WWTP series: laboratory protocol knowledge under time pressure. The near-unanimous correct response (adjusted 100%) confirms that sample preservation fundamentals are well-represented in current LLM training data. The designed acid trap caught no models — the first time in the series a trap has been universally avoided.

The behavioral findings are the primary contribution. A 3,000× token spread for the same correct answer — from Gemini 2.5 Flash's single character to Qwen Thinking's 3,009-token deliberation — illustrates that efficiency and correctness are independent dimensions. Neither the shortest nor the longest response is inherently better; they represent fundamentally different approaches to the same problem. Correctness is binary, but reasoning quality is multi-dimensional — and on this benchmark, the multi-dimensional view is where the signal lies.

R1's parsing pattern now appears in 5 of 7 benchmarks (71%), with a return to the <think> block format after two benchmarks of bold markdown. This is no longer an occasional artifact — it is a consistent behavioral trait that reliably produces scoring errors when evaluation relies on first-character extraction.

Gemma 1B's recovery from the most dangerous answer in the series (Confined Space: recommending unprotected entry) to a correct answer here reinforces a recurring observation: the smallest models are unpredictable rather than consistently wrong. This unpredictability itself may be the most important behavioral signal for safety-critical deployment decisions.

The series continues to evolve. Seven benchmarks, seven different cognitive dimensions, and still no model with a perfect record. The funnel at zero for three consecutive benchmarks suggests that consistent cross-domain performance remains beyond any single model's current capability.

Why was this question easy? The most likely explanation is that APHA Standard Methods for sample preservation is a widely referenced document in environmental science literature, and its core principles — cold storage, closed containers, dark conditions — appear frequently in training data. Questions grounded in well-established, widely-published standards may consistently produce high pass rates, while questions requiring trade-off reasoning or domain-specific judgment (Biogas at 14%, Walkway at 36%) appear to probe knowledge that is less densely represented.

7. Methodology Notes

Non-determinism: Single run per model. Results are indicative, not absolute. Future iterations may incorporate multi-run protocols to assess stability.

Single question: This report analyzes one question with 5 options. The random correct answer probability is 20%.

Laboratory context: The correct answer (D) is based on APHA Standard Methods for the Examination of Water and Wastewater. The scenario is derived from real operational conditions at a municipal WWTP in a hot climate.

Parsing: DeepSeek R1's response begins with <think>. The adjusted pass rate (100%) accounts for this. R1's parsing behavior is tracked as a systematic pattern across the series.

Token counts: Some models (Gemma 3 12B, 27B, 4B) reported 0 output tokens in API metrics despite producing visible responses. Token counts for these models are estimated from response character length and marked with ~.

API conditions: All models accessed via Kaggle Benchmarks platform API with default temperature settings.